Dimension reduction

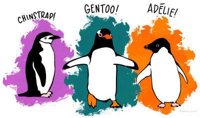
Master MIASHS

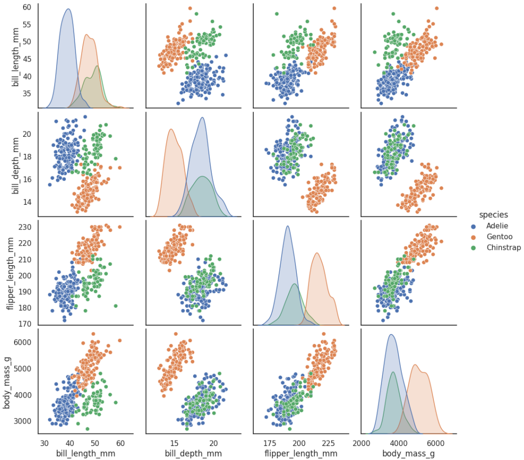
Paul Tresson

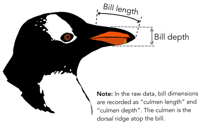
18/10/25

Introduction

Visualize what is happening in higher dimensions







Visualize what is happening in higher dimensions

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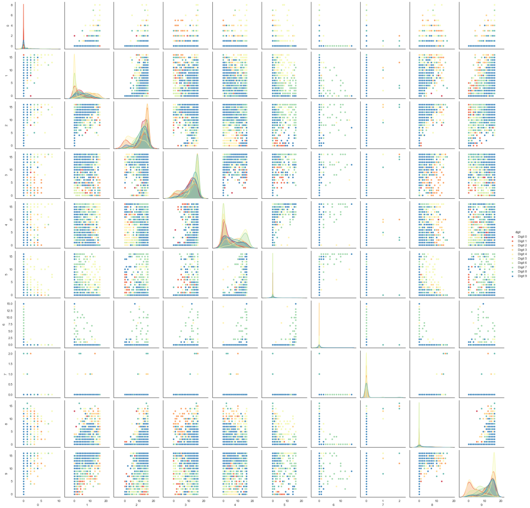

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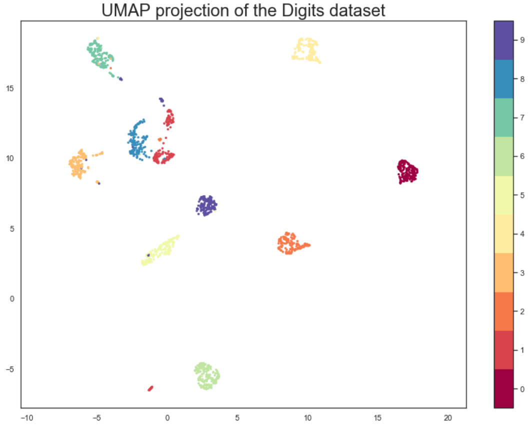
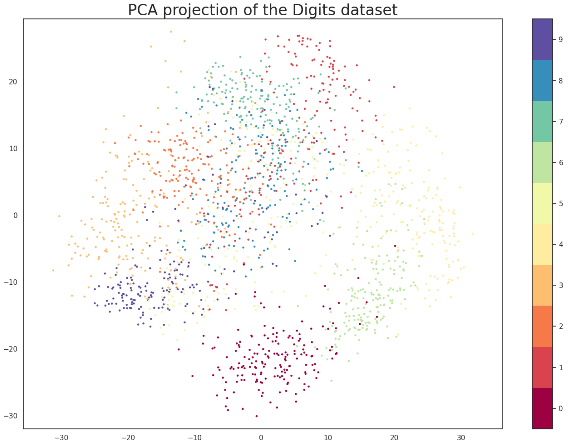
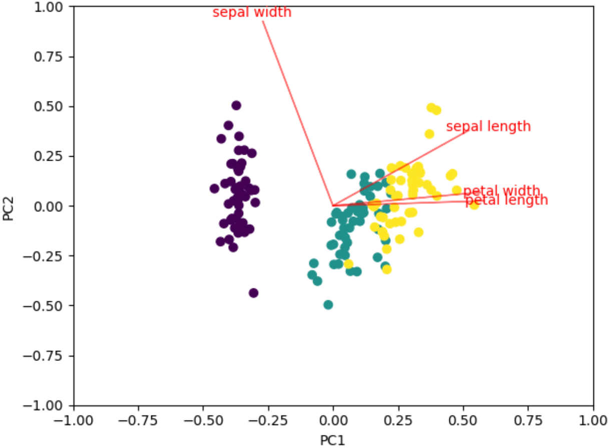


PCA


PCA projection of the Digits dataset
20
10
0
−10
−20
−30
−30
−20
−10
0
10
20
30
9
8
7
6
5
4
3
2
1
0

PCA interpretability



PCA advantages and drawbacks

Advantages

- fast

Drawbacks

PCA advantages and drawbacks

Advantages

- fast
- scales well

Drawbacks

PCA advantages and drawbacks

Advantages

- fast
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- explainatory

Drawbacks

PCA advantages and drawbacks

Advantages

- fast
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Drawbacks

- **not suited for non-linear data**

Non linear data

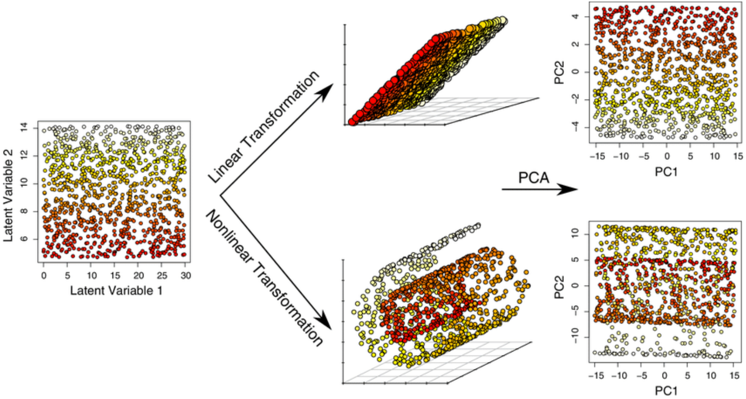

Figure from Du, 2019

Introduction

UMAP

How to work with non linear data

1. Find a good representation of the data in high dimension
2. Fit a good representation of in low dimension

Classics

- **SNE** (Stochastic Neighbor Embedding) Hinton and Roweis, 2002

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- **T-SNE** (T-distributed SNE) Van der Maaten and Hinton, 2008

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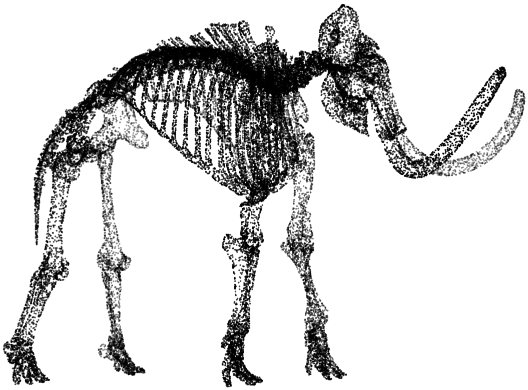
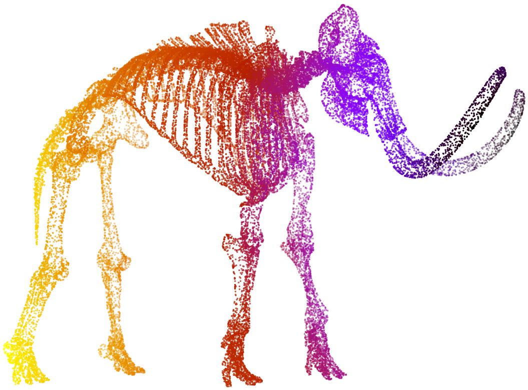
- **UMAP** (Uniform Manifold Approximation and Projection) McInnes et al., 2018

Mammoth example



Mammoth dataset by the Smithonian

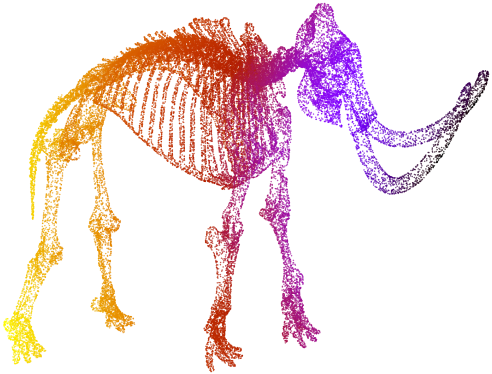
Mammoth example



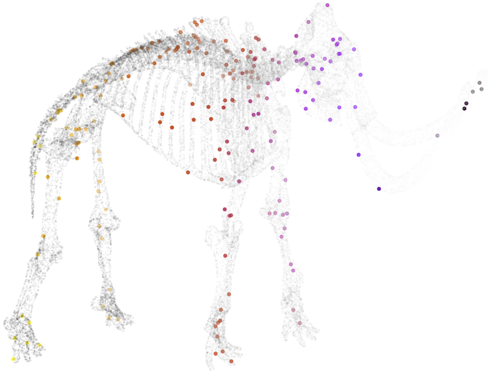
Mammoth dataset by the Smithonian

Mammoth example - PCA

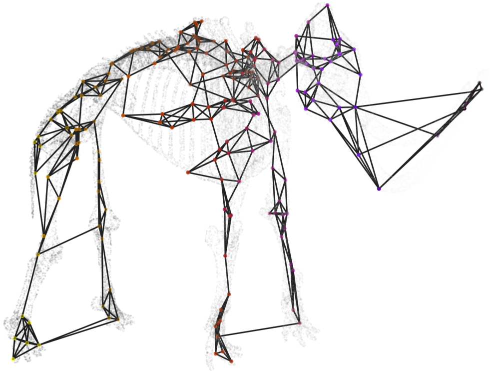
Finding high dimension graph



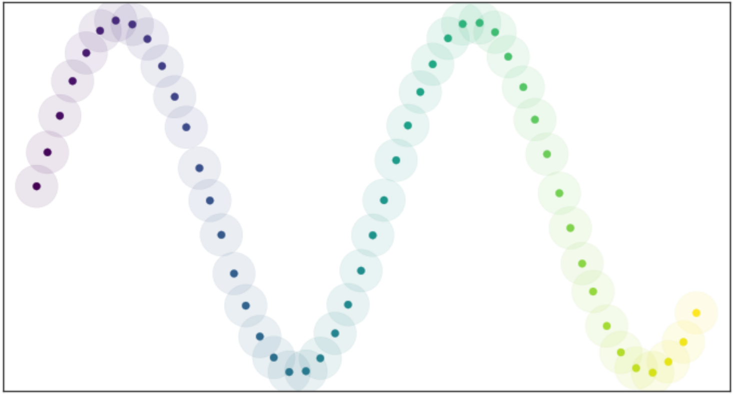
Finding high dimension graph



Finding high dimension graph

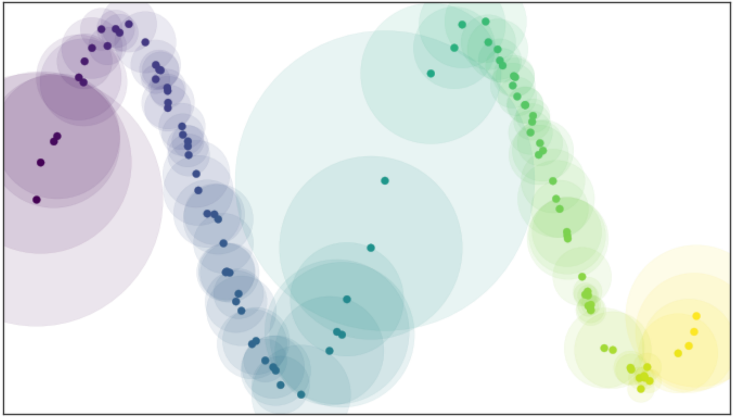


Uniform Manifold Approximation and Projection



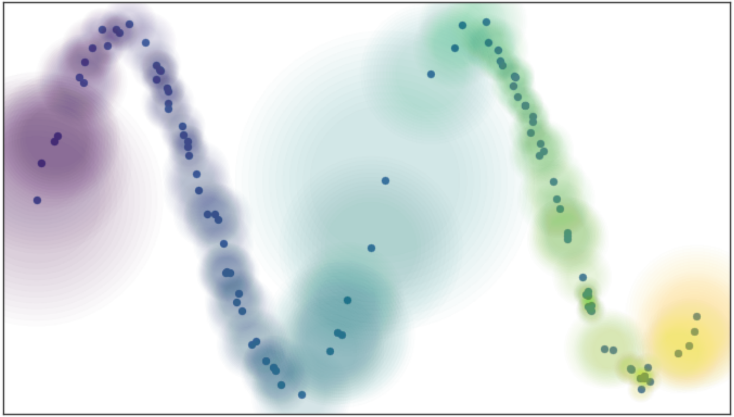
UMAP documentation

Uniform Manifold Approximation and Projection



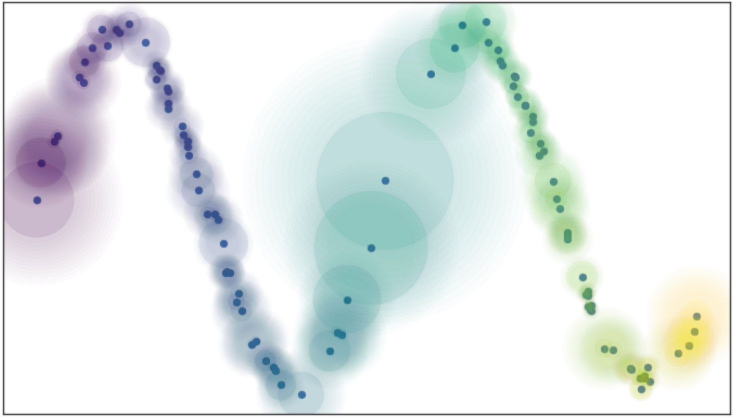
UMAP documentation

Uniform Manifold Approximation and Projection



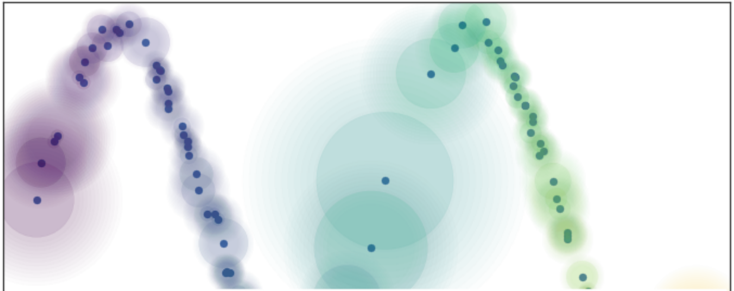
UMAP documentation

Uniform Manifold Approximation and Projection



UMAP documentation

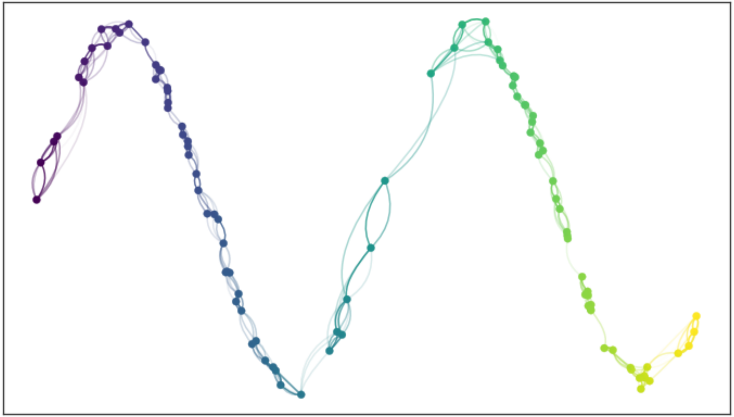
Uniform Manifold Approximation and Projection



$$\sum_{j=1}^{k} \exp\left(\frac{-\max(0, d(x_i, x_{i_j}) - \rho_i)}{\sigma_i}\right) = \log_2(k)$$

UMAP documentation

Uniform Manifold Approximation and Projection



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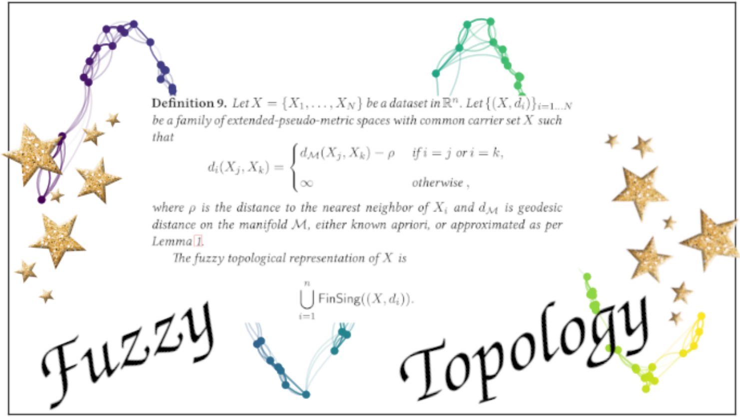

UMAP documentation

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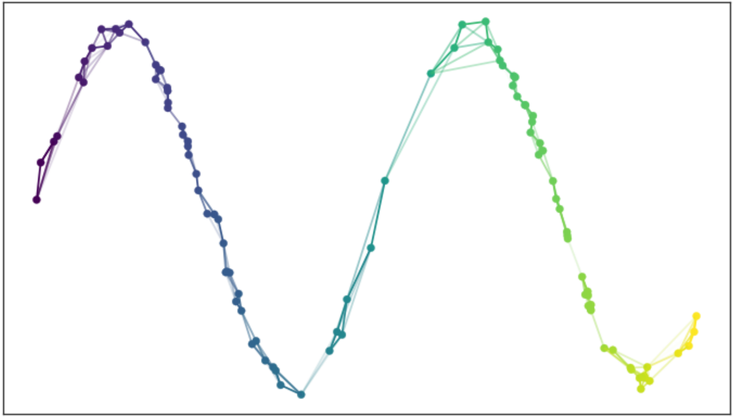
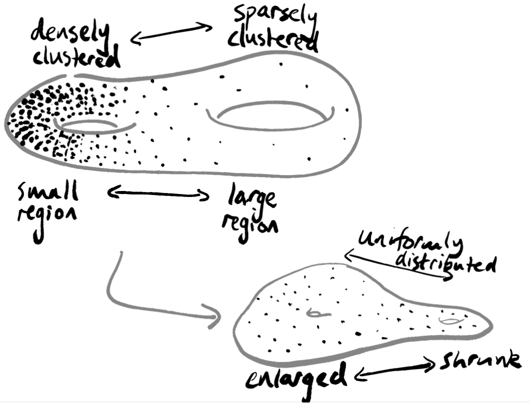
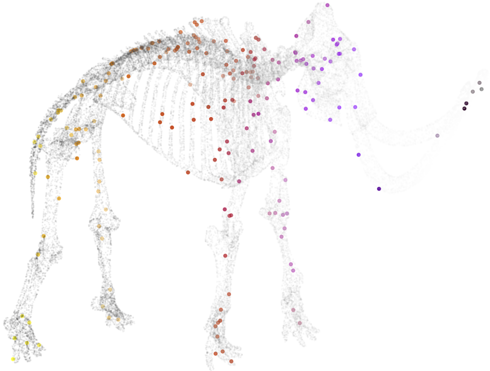


UMAP documentation

Uniform Manifold Approximation and Projection



Finding high dimension graph



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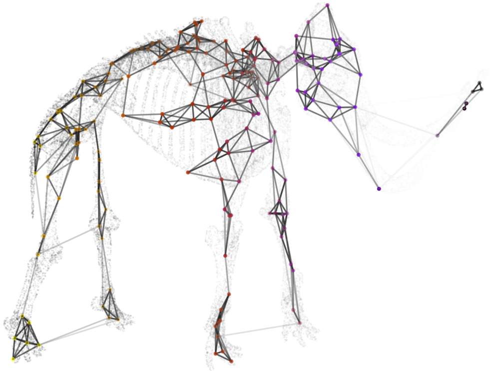
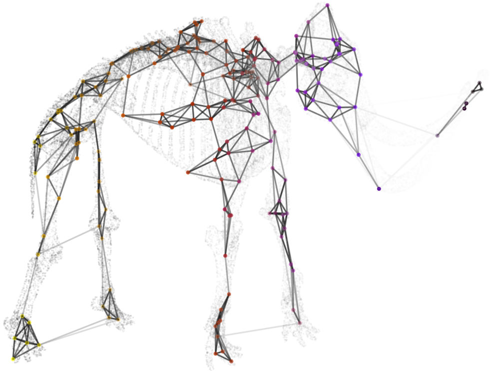


Fitting low dimensional representation



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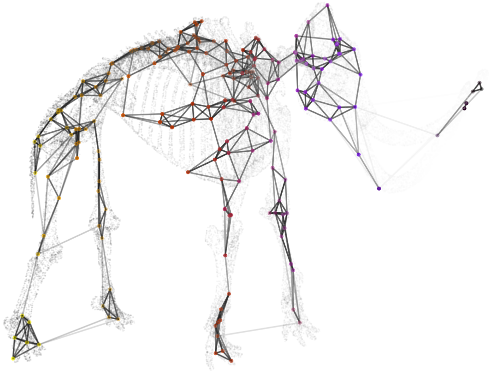

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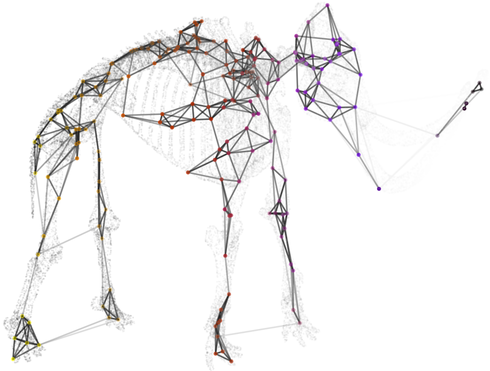
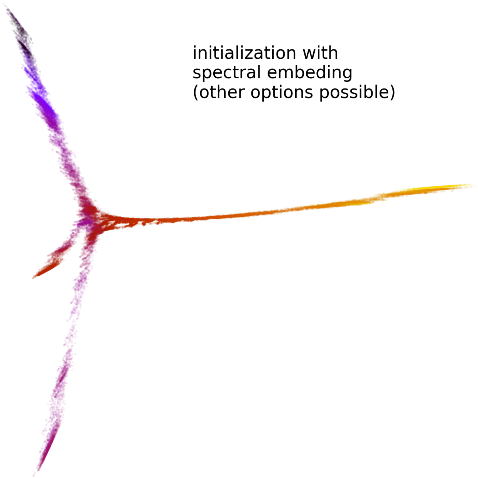
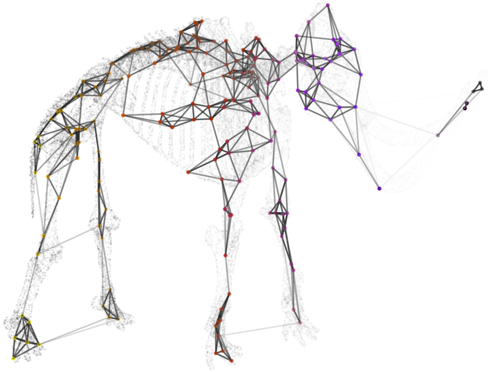
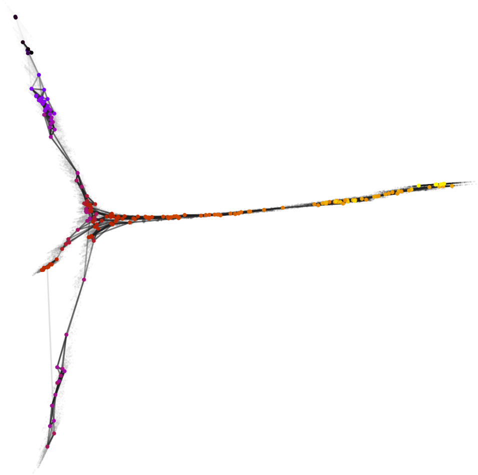




initialization with
spectral embeding
(other options possible)

Fitting low dimensional representation





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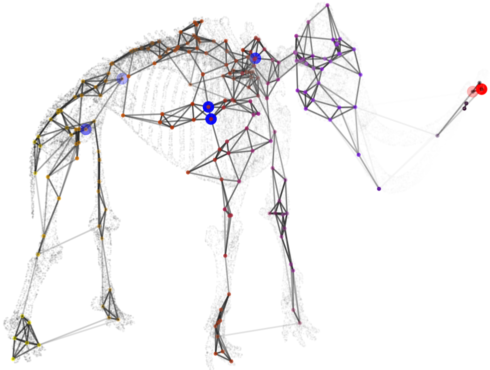
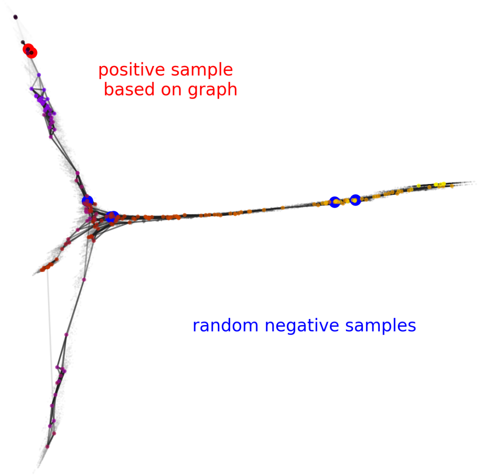




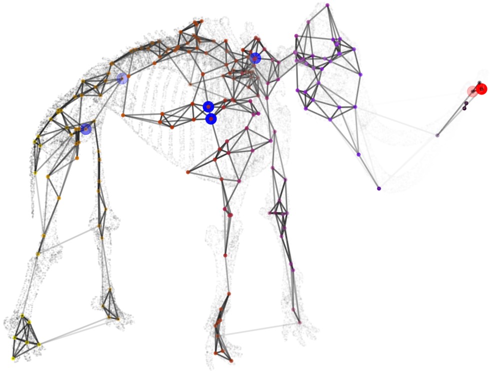
Fitting low dimensional representation

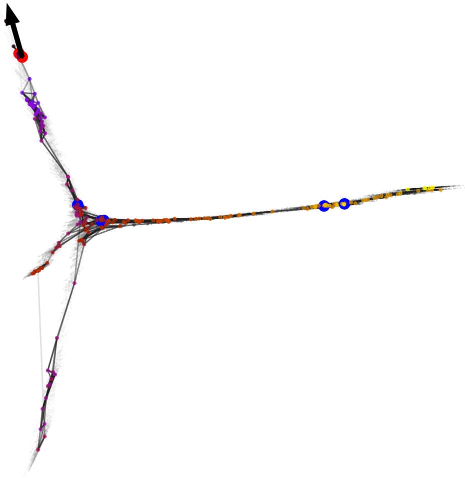
Cross-entropy loss counting positive and negative atraction forces

$$C_{UMAP} = \sum_{i \neq j} v_{ij} \log\left(\frac{v_{ij}}{w_{ij}}\right) + (1 - v_{ij}) \log\left(\frac{1 - v_{ij}}{1 - w_{ij}}\right) \tag{1}$$

Optimization with **Stochastic Gradient Descent**

Fitting low dimensional representation





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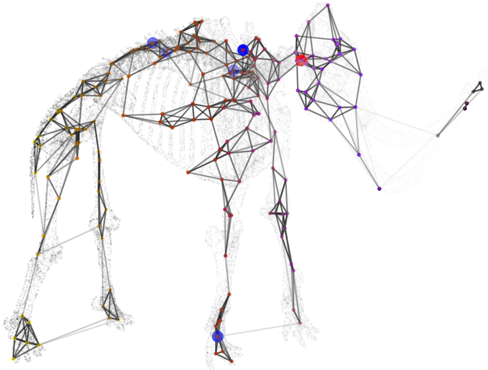
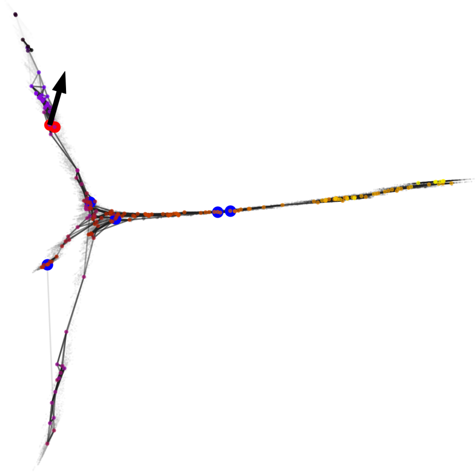
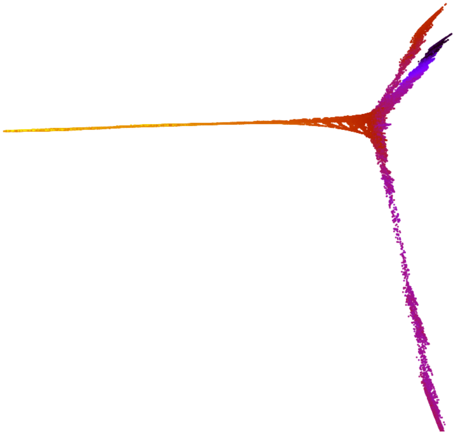




Optimization



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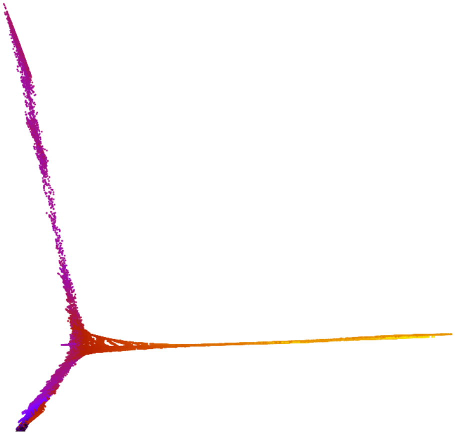
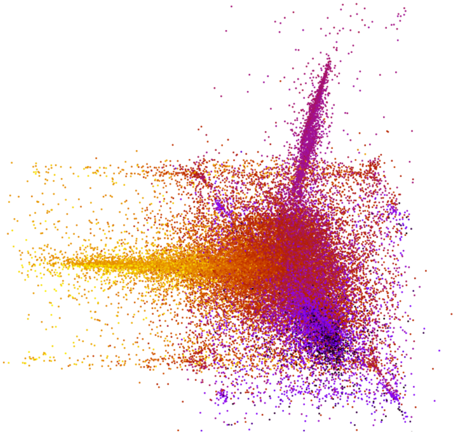
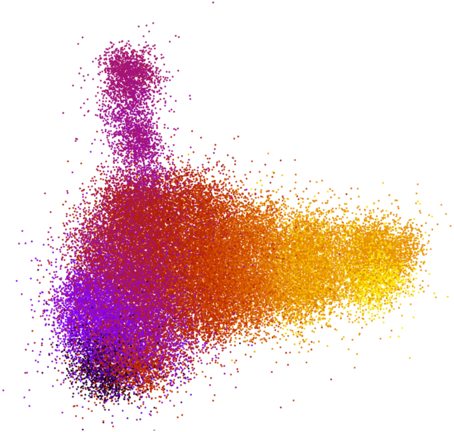


Optimization



Optimization

Optimization

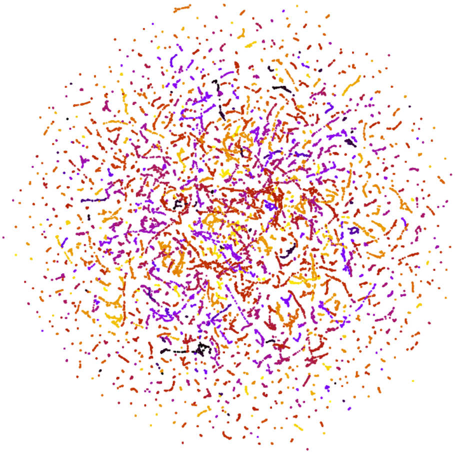


Optimization

Hyper-parameters

- number of neighbors
- min distance
- distance metric

Influence of the number of neighbors



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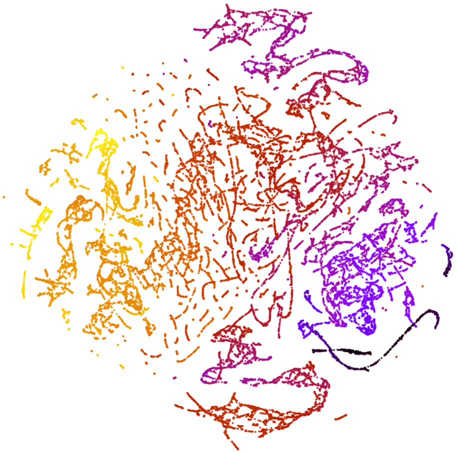

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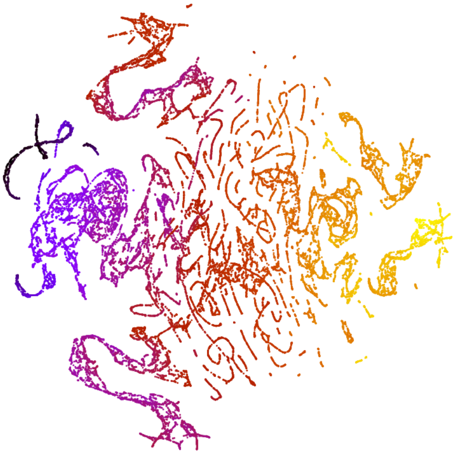

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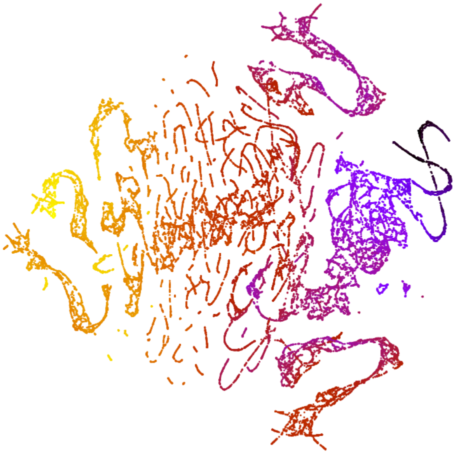

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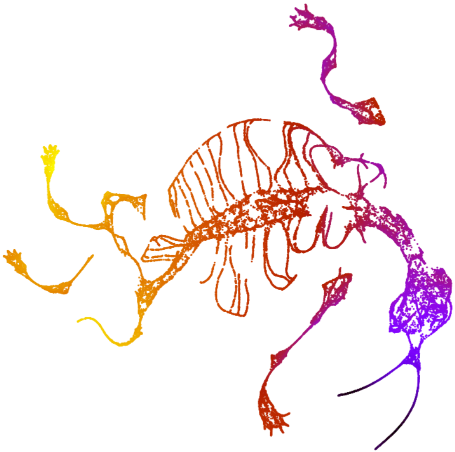

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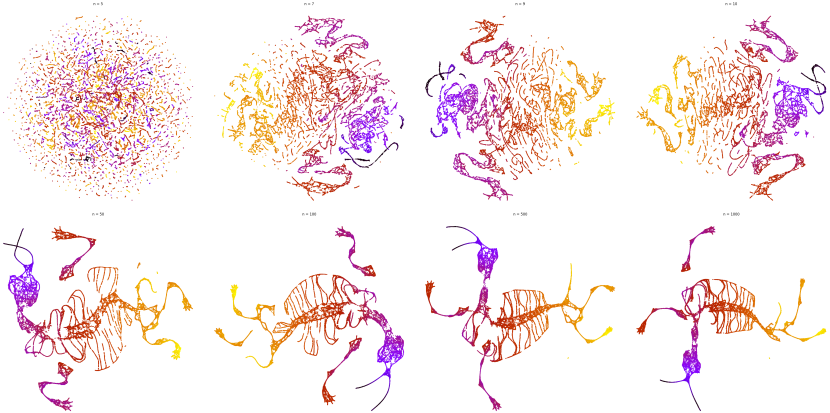


UMAP vs. T-SNE

UMAP

- graph theory, fuzzy logic

T-SNE

- probabilities

UMAP vs. T-SNE

UMAP

- graph theory, fuzzy logic
- determinist initialization

T-SNE

- probabilities
- random initialization

UMAP vs. T-SNE

UMAP

- graph theory, fuzzy logic
- determinist initialization
- log_2 similarity scores

T-SNE

- probabilities
- random initialization
- gaussian distribution similarity

UMAP vs. T-SNE

UMAP

- graph theory, fuzzy logic
- determinist initialization
- *log*$_2$ similarity scores
- update by pairs

$\rightarrow$ scales well for large datasets

T-SNE

- probabilities
- random initialization
- gaussian distribution similarity
- update all points

scales less

Scaling

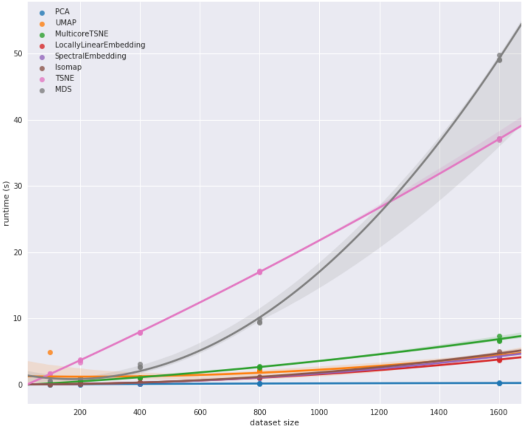
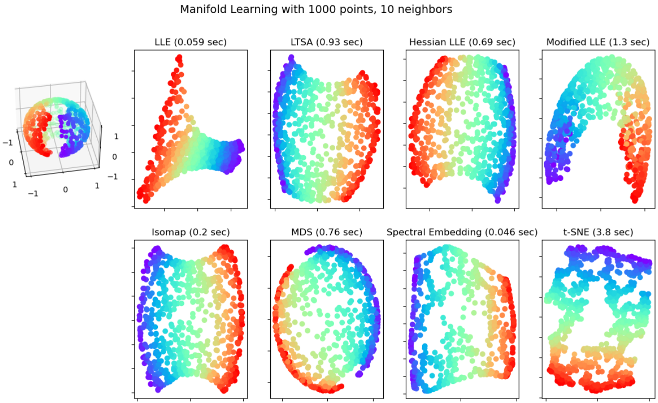


Figure from umap-learn documentation

A lot of options !



Useful ressources

- `scikit-learn` docs !
- deepia
- StatQuest

Thanks for your attention !

Let's practice !

Du, Trina Y (2019). **"Dimensionality reduction techniques for visualizing morphometric data: Comparing principal component analysis to nonlinear methods".** In: *Evolutionary Biology* 46.1, pp. 106–121.

Hinton, Geoffrey E and Sam Roweis (2002). **"Stochastic neighbor embedding".** In: *Advances in neural information processing systems* 15.

McInnes, Leland, John Healy, and James Melville (2018). **"Umap: Uniform manifold approximation and projection for dimension reduction".** In: *arXiv preprint arXiv:1802.03426*.

Van der Maaten, Laurens and Geoffrey Hinton (2008). **"Visualizing data using t-SNE.".** In: *Journal of machine learning research* 9.11.